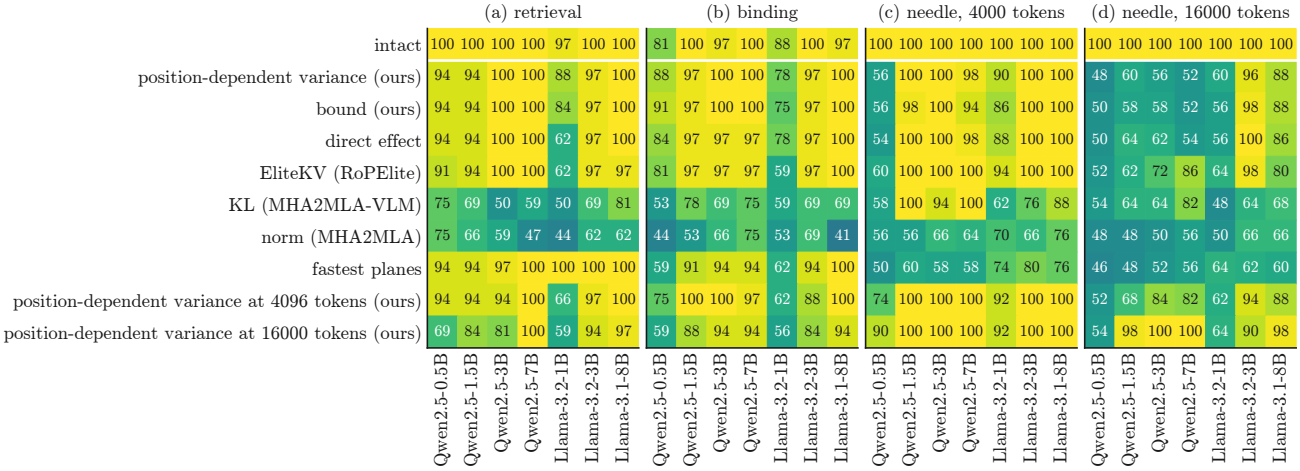
(a) retrieval
(b) binding
(c) needle, 4000 tokens
(d) needle, 16000 tokens
intact 100 100 100 100 97 100 100 81 100 97 100 88 100 97 100 100 100 100 100 100 100 100 100 100 100 100 100
position-dependent variance (ours) 94 94 100 100 88 97 100 88 97 100 100 78 97 100 56 100 100 98 90 100 100 48 60 56 52 60 96 88
bound (ours) 94 94 100 100 84 97 100 91 97 100 100 75 97 100 56 98 100 94 86 100 100 50 58 58 52 56 98 88
direct effect 94 94 100 100 62 97 100 84 97 97 97 78 97 100 54 100 100 98 88 100 100 50 64 62 54 56 100 86
EliteKV (RoPElite) 91 94 100 100 62 97 97 81 97 97 97 59 97 100 60 100 100 100 94 100 100 52 62 72 86 64 98 80
KL (MHA2MLA-VLM) 75 69 50 59 50 69 81 53 78 69 75 59 69 69 58 100 94 100 62 76 88 54 64 64 82 48 64 68
norm (MHA2MLA) 75 66 59 47 44 62 62 44 53 66 75 53 69 41 56 56 66 64 70 66 76 48 48 50 56 50 66 66
fastest planes 94 94 97 100 100 100 100 59 91 94 94 62 94 100 50 60 58 58 74 80 76 46 48 52 56 64 62 60
position-dependent variance at 4096 tokens (ours) 94 94 94 100 66 97 100 75 100 100 97 62 88 100 74 100 100 100 92 100 100 52 68 84 82 62 94 88
position-dependent variance at 16000 tokens (ours) 69 84 81 100 59 94 97 59 88 94 94 56 84 94 90 100 100 100 92 100 100 54 98 100 100 64 90 98
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-1B
Llama-3.2-3B
Llama-3.1-8B